AI Security Use Cases

SECAI+ PREP LESSON 5.1.3

AI tools are transforming how security professionals work from reviewing code quality to threat modeling and incident analysis. This lesson features an expert discussion on the most impactful AI security use cases shaping modern cybersecurity practice.

Code Quality Assurance

AI goes far beyond simply generating code — it actively analyzes existing scripts and custom-written tools to identify improvements, inefficiencies, and best practices.

Review Custom Code

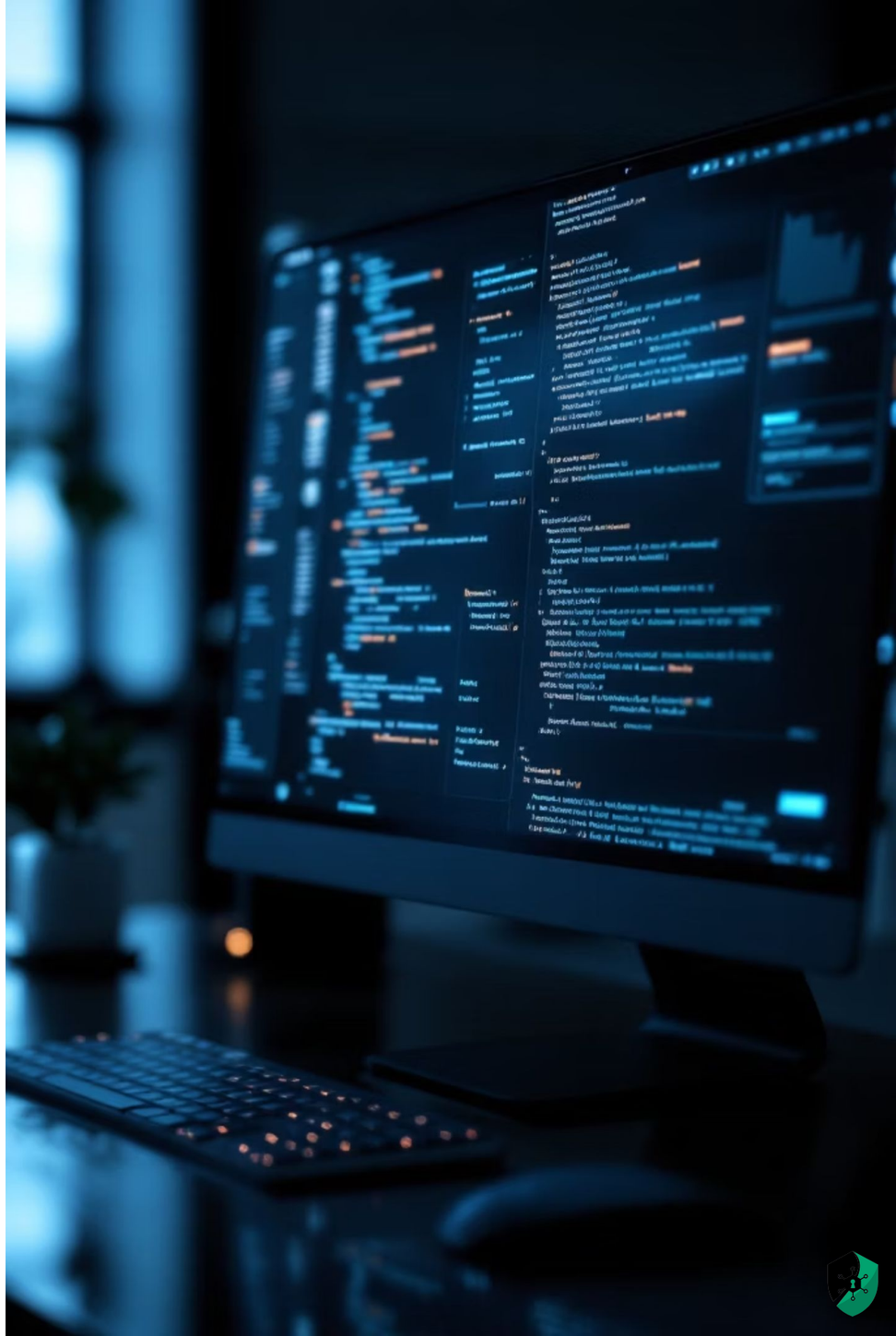
AI can review code written by technicians for quality and efficiency, catching issues that manual review might miss.

Trained on Millions of Lines

Trained on vast codebases, AI recognizes best practices across languages and frameworks with expert-level pattern recognition.

Optimization Suggestions

AI can suggest achieving the same result in fewer lines of code, reducing complexity and improving maintainability.



Threat Modeling with TTPs

TACTICS · TECHNIQUES · PROCEDURES

AI tools help security teams analyze the **tactics, techniques, and procedures (TTPs)** of known threat organizations dramatically accelerating the identification and classification of adversaries.

→ Classify Incidents

Classifies incidents based on historical attack patterns and known threat actor behaviors.

→ Identify Threat Groups

Identifies threat groups by IP ranges, malware types, and specific attack methods.

→ Faster Than Manual Research

Reaches analysis conclusions much faster than traditional human-driven research workflows.

Why TTPs Matter

TTPs represent the behavioral fingerprint of a threat actor. By mapping observed activity to known TTPs, AI can attribute attacks to specific groups and predict next moves giving defenders a critical edge.

API Integrations for Holistic Analysis

AI threat tools leverage API integrations to pull data from multiple security sources simultaneously, delivering a cohesive, big-picture view of what is occurring across the entire environment.

Vulnerability Assessment

Integrates with tools like **Nessus** to surface known vulnerabilities across assets.

Network Scanning

Pulls data from **Nmap** and **Angry IP Scanner** for live network topology and host discovery.

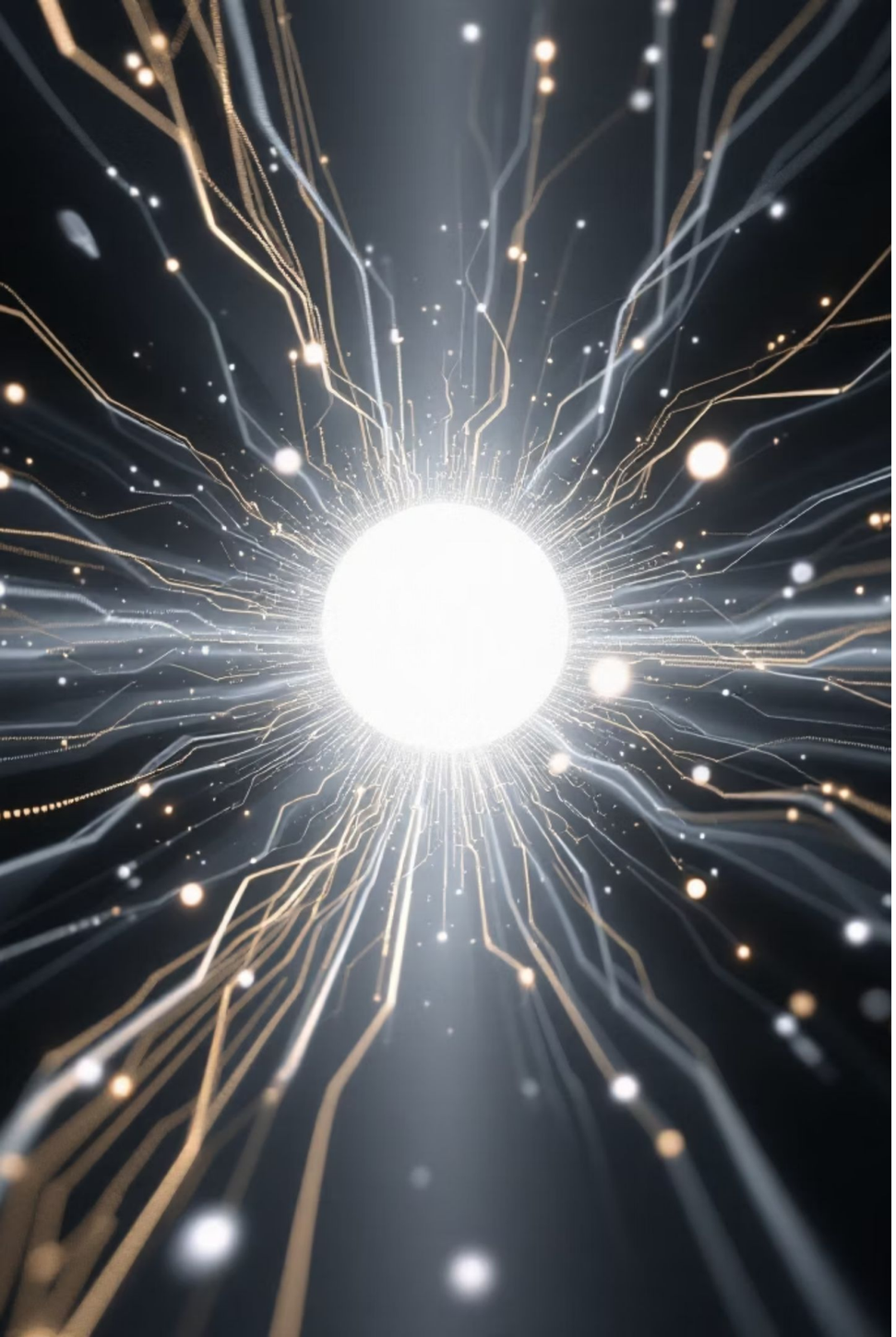
SIEM Alerts

Ingests alerts from platforms like **Splunk** to correlate events across the environment.

Router & Switch Logs

Analyzes infrastructure logs to detect anomalous traffic patterns and lateral movement.

By unifying these data streams, AI delivers a single, correlated view of threats eliminating the blind spots that come from analyzing tools in isolation.



Summarization and Synthesization

What AI Analyzes

AI rapidly processes **millions of documents and reports**, synthesizing inputs from all integrated sources to identify the most likely threat scenario — a task that would take human analysts days or weeks.

1. Ingest millions of data points
2. Synthesize across all sources
3. Produce actionable output reports

What AI Produces

Output reports explain **what is happening and why**, giving analysts immediate context without requiring them to manually correlate data.

🔍 Indicators of Compromise

Highlights specific IoCs and links them directly to known threat actors and campaigns.

📋 Threat Attribution

Connects observed behaviors to threat actor profiles using historical TTP data.

Balancing Big Picture and Details

Effective incident response requires both **rapid high-level decision-making** and **deep forensic analysis**. AI enables security teams to operate at both levels simultaneously.

High-Level Summaries

Provides rapid summaries for immediate decision-making, enabling teams to act without waiting for full analysis.

Preserved Evidence & IoCs

Retains specific evidence and indicators of compromise for thorough post-incident analysis and forensics.

Accelerated Response

Teams respond immediately to active threats while detailed review is queued for later, dramatically speeding up the overall incident response process.

Containment Actions

Isolate systems and block malicious activity quickly.

Forensic Analysis

Deep IoC review and timeline reconstruction.

Immediate Response

AI-driven summary for rapid decisions and containment.

Lessons Learned

Document findings and update defenses.

Key Takeaways

LESSON 5.1.3 SUMMARY

01

Code Quality Review

AI excels at reviewing custom security scripts for quality, efficiency, and best practices going beyond code generation to active improvement.

02

TTPs-Based Threat Modeling

TTPs-based analysis helps identify threat actors faster than manual research by matching behaviors to known adversary profiles.

03

API-Driven Holistic Analysis

API integrations allow AI to analyze data holistically across Nessus, Nmap, Splunk, and infrastructure logs for a unified threat picture.

04

Rapid Summarization

Summarization saves analyst time and accelerates incident response by synthesizing millions of data points into actionable reports.

05

Evidence Preservation

Details and IoCs are preserved for post-incident review and forensics, ensuring nothing is lost in the rush to respond.

Bottom Line: AI doesn't replace security analysts — it amplifies their capabilities, enabling faster decisions, deeper analysis, and more effective defense across the entire threat lifecycle.